$$r = \frac{N\sum dxdy - (\sum dx)(\sum dy)}{\sqrt{N\sum dx^2 - (\sum dx)^2}\sqrt{N\sum dy^2 - (\sum dy)^2}}$$

| S.No | X | Y | (X-A)=dx | (Y-A)=dy | dx2 | dy2 | Dxdy |
|---|---|---|---|---|---|---|---|
| 1 | 10 | 19 | -6 | -8 | 36 | 64 | 48 |
| 2 | 12 | 22 | -4 | -5 | 16 | 25 | 20 |
| 3 | 13 | 26 | -3 | -1 | 9 | 1 | 3 |
| 4 | 16 | 27 | 0 | 0 | 0 | 0 | 0 |
| 5 | 17 | 29 | 1 | 2 | 1 | 4 | 2 |
| 6 | 20 | 33 | 4 | 6 | 16 | 36 | 24 |
| 7 | 25 | 37 | 9 | 10 | 81 | 100 | 90 |
| N=7 | ΣX=113 | ΣY=193 | Σ(X-A)=1 | Σ(Y-A)=4 | $\Sigma dx^2$=159 | $\Sigma dy^2$=230 | Σdxdy=187 |

$$\bar{x} = \frac{\Sigma X}{N} = \frac{113}{7} = 16\frac{1}{7}$$

$$\bar{y} = \frac{\Sigma Y}{N} = \frac{193}{7} = 27\frac{4}{7}$$

Take the assumed values A = 16 & B = 27 therefore dx = X – A $\Rightarrow$ X – 16 and

dy = Y- A $\Rightarrow$ Y – 27

$$\therefore r = \frac{N\sum dxdy - (\sum dx)(\sum dy)}{\sqrt{N\sum dx^2 - (\sum dx)^2}\sqrt{N\sum dy^2 - (\sum dy)^2}}$$

$$= \frac{7\times187 - 1\times4}{\sqrt{7\times159 - (1)^2}\sqrt{7\times230 - (4)^2}}$$

$$= \frac{1309 - 4}{\sqrt{1112}\sqrt{1610 - 16}}$$

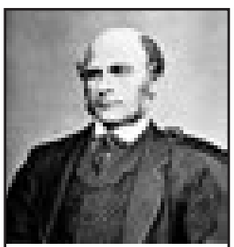
$$= \frac{1305}{\sqrt{1112}\sqrt{1594}} = \frac{1305}{(33.34)(39.92)}$$

$$= \frac{1305}{1330.9} = 0.9865 \qquad r = 0.986$$

There exists a positive high correlation between X and Y

## 12.13 Regression

### Evolution of Regression


Francis Galton

The term 'Regression' was first coined and used in 1877 by Francis Galton while studying the relationship between the height of fathers and sons. The average height of children born of parents of a given height tended to move or "regress" toward the average height in the population as a whole. Galton's law of universal regression was confirmed by his friend Karl Pearson, who collected more than a thousand records of heights of members of family groups. The literal meaning of the word "regression" is **"Stepping back towards the average".**

Regression is the study of the relationship between the variables. If Y is the dependent variable and X is independent variable, the linear relationship between the variable is called the regression equation of Y on X, The



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